

Assignment-1 (SDLC)

SDLC Overview

Infographic Description : Software Development Life Cycle (SDLC) Phases.

i) Requirements Gathering

- Purpose : Identify the need and expectations of users and stakeholders.
- Importance : forms the foundation for all future development.
- Outputs :- Requirement specification documents, use cases.

ii) Design

- Purpose : Define system architecture, components, modules, interfaces.
- Importance : Provides a blueprint that guides implementation.
- Outputs : Design documents, ER diagrams, UML models.

iii) Implementation (Coding)

- Purpose : Convert design into functional code.
- Importance : core development phase where actual product is built.
- Outputs : Source code, code repositories.

iv) Testing

- Purpose : Identify and fix bugs; ensure software meets requirements.
- Importance : Guarantees quality and reliability.
- Outputs : Test plans, test cases, bug reports.

→ Deployment & Maintenance

- Purpose: Release the product to users and perform ongoing updates.
- Importance: Ensures the software stays usable and relevant.
- Outputs: Deployment guides, release notes, maintenance logs.

Interconnection: Each phase flows into the next, with feedback loops. For instance, testing may uncover design issues; deployment may trigger new requirements.

Visual suggestion: Arrange the phases in a circular or waterfall flowchart with arrows and color-coded blocks indicating iterative loops.